NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Osteopathic Considerations in Gynecology

Reem Abu-Sbaih, D.O.
Professor
Department of OMM
rabusbai@nyit.edu
Rm. 134 Serota



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

“One of the most important branches taught in the American School of Osteopathy– Gynaecology…”

AT Still p.202 Philosophy and Mechanical Principles of Osteopathy



ORIGINAL CLASS IN OSTEOPATHY, 1892-93

(Photo courtesy Still National Osteopathic Museum, Kirksville)

Session Objectives

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

1. Understand how **anatomic/structural changes & wounds impact the physiologic function or dysfunction** that may produce/promulgate **somatic dysfunction** (SD). Compare and contrast internal vs. external considerations in health and disease in Gynecology - highlight structural considerations at different life stages and how osteopaths can be of service.
2. Correlate importance of perfect **Circulation** in health and disease of gynecologic concerns - Understand the **role of thorax, diaphragm & lymphatics** for the female patient whether for reproductive organs or otherwise. Understand the osteopathic physicians' role and responsibility with regard to "the want of supply and the burden of dead deposits"
3. Describe the importance of healthy balance of **Nervous system** in health & disease of gynecologic concerns. Understand importance of the **autonomic nervous system** (and "**facilitation"** of related segments) with respect to reproductive function and the whole patient. Discuss what **spinal levels or craniosacral levels** are involved with **autonomics & circulation** to/ from the GU system**
4. Discuss Osteopathic Considerations of Common Gynecologic Complaints.
5. Review and Describe the genitourinary **Chapman's points**

Examination

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- "… to do justice in gynaecological service, (one) must be a workhand in the navy-yard of life, and must examine the whole vessel when she comes in to the dock for repairs. It is not necessary to look at the bottom of the ship all the time. Look at all parts with equal energy."

  -Andrew Taylor Still

Philosophy and Mechanical Principles of Osteopathy (P&MPO) p 211-212

Where to start looking for Health?

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Diaphragm (nerve and arterial supply)
- Pancreas, liver, GB, spleen:
  - Derangement related to
    - Dyspepsia, dropsy, enlarged spleen, engorged liver, cancer, gall-stones, skin eruptions, change of color, constipation, inflammatory diseases, ulceration of the stomach, bowels, kidneys and uterus.
- Remove embarrassments from blood and nerve supply which could be caused by: strains, jars, nervous shocks, or wounds produced by change of season, climate, physical injuries of all kinds, be they great or small.

Begin at:

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

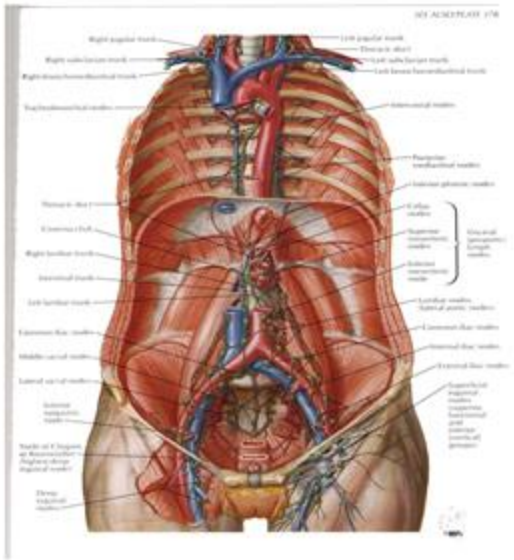
- The bones and know them, their uses and their places
- Then the binding ligaments…
- In particular her spine and pelvis…
- Search until we know what effect would be produced on any organ of the chest or abdomen by an anterior, posterior or lateral change of the atlas, axis or other bones of the neck;
- What effect on the heart would follow the changed position of a rib…?
- T1-T12, lumbar, sacrum innominates, bc of a slip or change from the normal. Would it send more or less blood to the brain?...

Considerations Continued

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Lungs to suffer loss of power purifying blood and then a failure in the power of the heart would appear,…
- Then a stagnation … liver, spleen pancreas….
- Ad infinitum…congestion and sluggish action in the organs upon which a healthy womb has to depend for its vitality and functioning….”

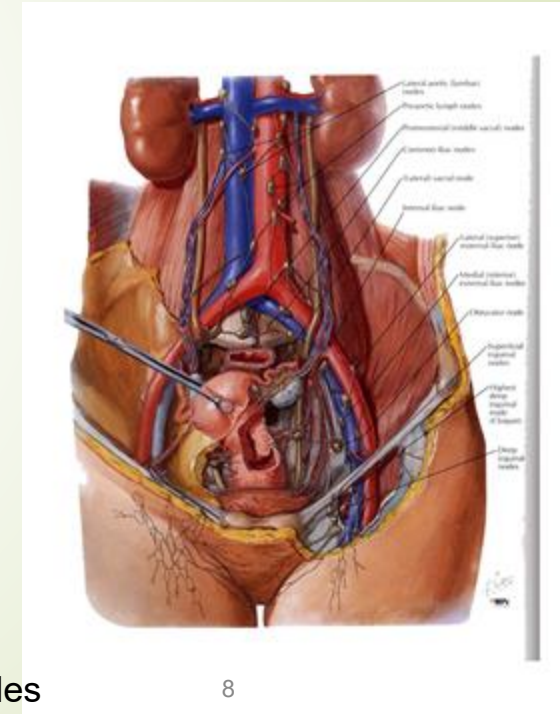


More thoughts from Dr Still P&MPO text

Stop and Think

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- "Can an organ be normal and be starved?"
- "Can an organ be normal and be twisted when it in naturally straight?"
- "Can an organ do its part with other organs pressing on it too heavily?"

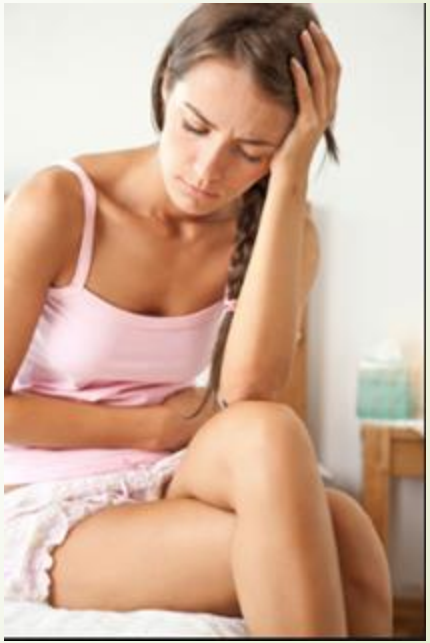


Gynecologic Considerations

NEW YORK INSTITUTE OF TECHNOLOGY

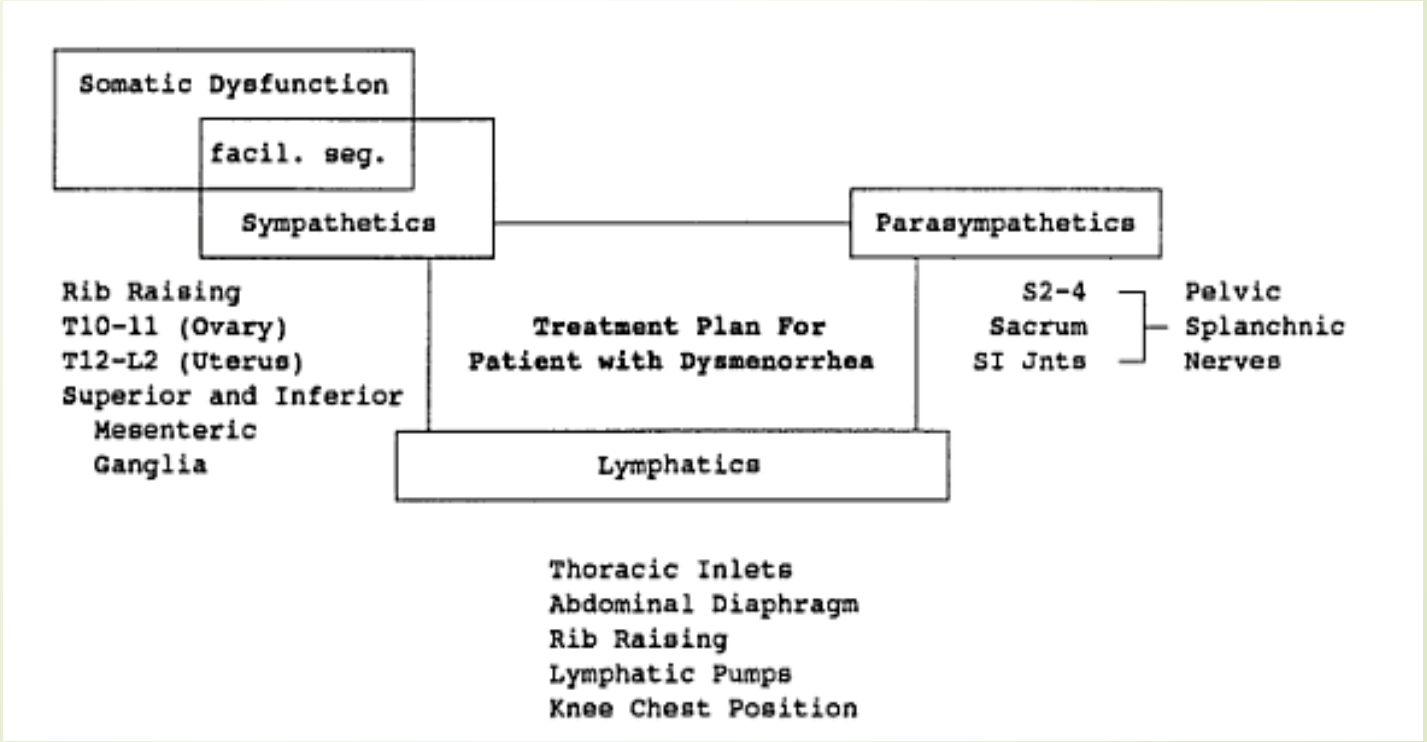
College of Osteopathic Medicine

- Spine (Exaggerated curves: A-P, lateral)
- Pelvic Floor Dysfunction
- Dysmenorrhea (primary or secondary)
- Pre-menstrual syndrome
- Breast Heaviness/ Pain
- Pelvic Congestion
- Myomata (Fibroids)
- Leucorrhea
- Pelvic Inflammatory Disease
- Endometriosis
- Adhesions
- Ovarian cysts/ remnants
- Tumefaction
- Dyspareunia
- Infertility
- Breast



Gynecological Considerations (Kuchera)

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine



Structural Concepts from Childhood to Womanhood

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine



Image: https://media4.picsearch.com/is?s6DS4q1z_WxJ9cNovMERJ0Xso-Dpsn4XB0pF9HGXqSI&height=341

To find health must be the object of the physician.
Anyone can find disease. – A.T.S.

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

"You must know what a healthy woman is before you can think and act wisely with the woman who has lost her health, say of sight… affecting mind… bowels… womb… heart, lungs… and all parts of a perfect woman. It is the perfect woman I want to present to your mind. The first thought of the successful osteopath is perfection, and he must place in mind perfection of form and function of the woman and keep that picture bright in his mind's eye all the time or he will be a failure as a gynaecologist all his days"

-A.T.S.



- https://media4.picsearch.com/is?ML6LYp2FrfuRaOQa71kjI8COR84t3LvQRv3IsjhBmb4&height=227
- P&MPO reading

From Childhood



W YORK INSTITUTE
TECHNOLOGY

lege of Osteopathic
dicine

"Full of blood and full of life, quick in motion, active in mind, able to answer and act to all the functions of life"

-Dr Still P&MPO



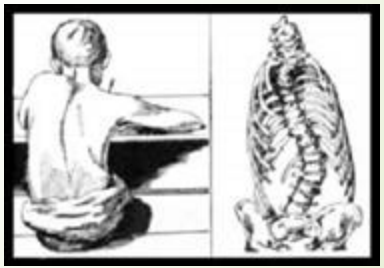
https://media2.picsearch.com/is?_9xII48DEUFu4LKjwblDYVfbYoK8fsjxloQmCUdshel&height=312;
https://media3.picsearch.com/is?eCO6cUP-HVtTOWP1pJ6rXzrP05_0d7qAGR07ITCrsiA&height=227

Repetitive Strain Patterns

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine







https://media3.picsearch.com/is?2u9gkLoQQQLDIYIb__YGPJDAuxwXPjZAmOQq9Wu9-k8&height=269
https://media5.picsearch.com/is?ru4zm43NoC05qHUd83rRiMdcY1qSn5VoYa08qwLIXA&height=173
https://media2.picsearch.com/is?KOSzc8gJIm1zI4_p2IFp-rQRpxixCTj9Gi9vmaVTQqg&height=175

Adolescence

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine









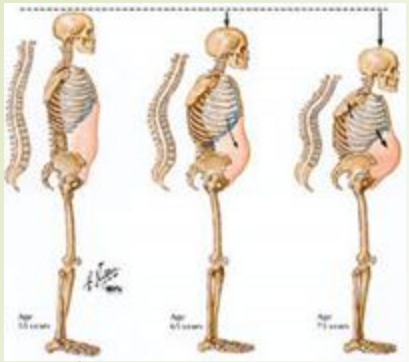
https://media2.picsearch.com/is?ZJ9YpQz_IHiqbguLjbV9p5q5lHTTV85z3-ZdJdjEMY&height=341

https://media4.picsearch.com/is?pS_h1v4sFuPxYWoUkwDhyKrC66FOtHNMO42HkBMLJzY&height=193

Effects of Wounds

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Congestion following changes of weather is a wound….
- Or injury to womb hurt in a fall, strain, jar, or shock produced by change of clothing …
- Severity of the shock is shown in the organs by soreness of the abdomen/ tenderness feeling of weight in pelvis followed sooner or later by bladder trouble, external/ internal reproductive issues

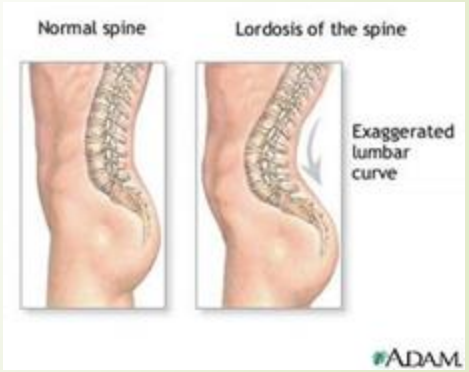




NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine





https://media2.picsearch.com/is?J6BAY-32zVTdgM1jZxrgna3OrUa7irtOXTzF_15ZKmY&height=248
https://media3.picsearch.com/is?1WvsSsonZ75gRufSwb88Y22LZR0bQz44FvzgA2-gc60&height=213
https://media1.picsearch.com/is?ocek-Ar0aurw6WhoAlPiJx7DI6VYqZt_RbxPuWwh7xY&height=189
Image: https://media3.picsearch.com/is?gtNW_jPZGM2MQTViSaRQB7wT3FFHL4KY-LKeDE5OmAk&height=256

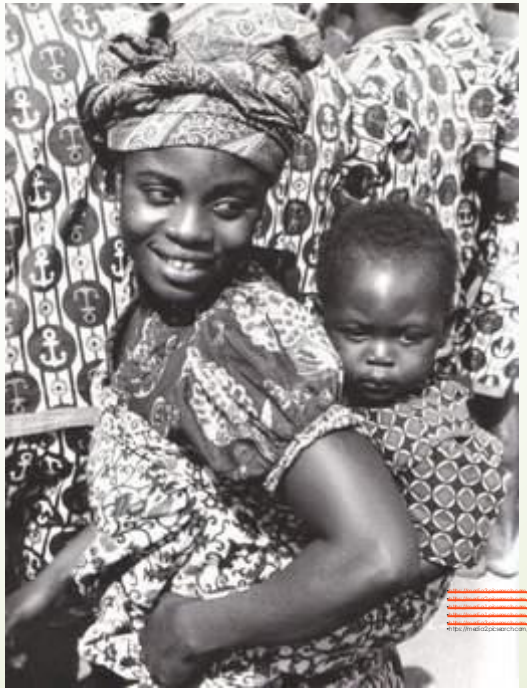
To Womanhood – Keep picture of Health

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine













NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Physiologic Changes

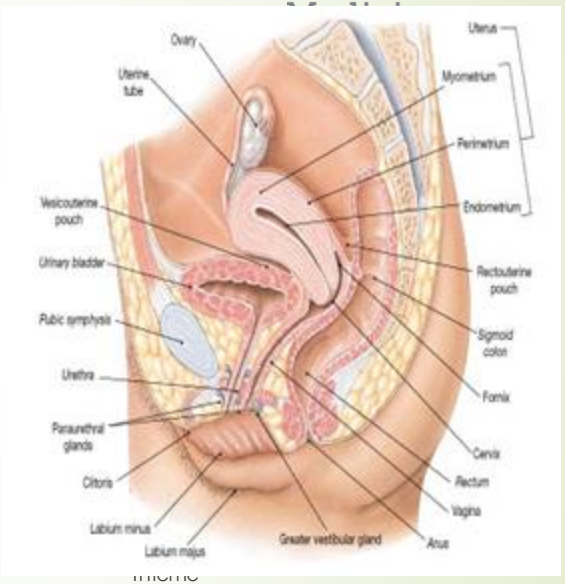
Structure and Function are Interrelated

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic

Consider Impact of:

- Wounds
- Crowding
- Postural issues
- Nutrition
- Exercise or lack thereof
- Surgical scars/ adhesions
- Scoliosis
- Sympathotonia
- Inflammation
- T.A.R.T.
- Gestation
- External stressors- tight clothing
- Psychological or physical trauma
- Impeded/Hindered blood supply
- Impeded/Hindered drainage



*Importance of Thoraco-Lumbar (T-L) Junction

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- An area of MSK Strain

- **Good motion T-L important for proper Diaphragmatic excursion!**
  - Crus of posterior diaphragm attachment L1-3 or L4
  - Think systemically: lymphatic and venous drainage

- **T10-L2** = SNS for **bowel, uterus and ovaries/ adnexa, bladder and ureters**, adrenals and renal
  - Any facilitation at these somatic segments by one organ can affect other organs supplied by the same segments

- *Exacerbation of facilitation of this area is common with MSK TART changes*

Impact of Wounds or TART on Ovaries

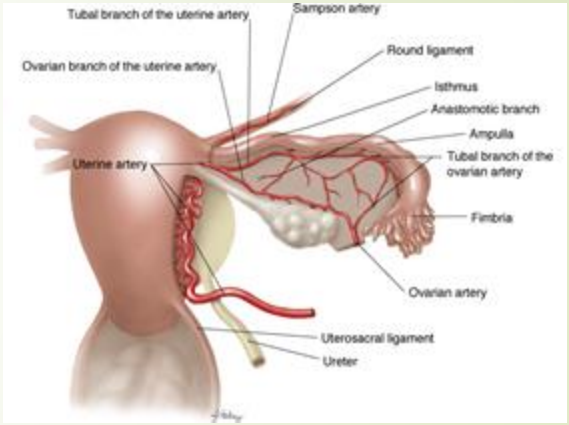
NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Arterial Supply: Ovarian artery
from aorta ~level of **L2**, Anastomose with branches of Uterine artery.

Venous Drainage:
Right: ovarian vein to the IVC
Left: to renal vein

Lymph Drainage: peri-aortic

SNS: T10-11
PSNS: Vagus



"*Mittelschmerz" referred through afferents along T10 dermatome – periumbilical area*

NOTE: Ovary often close to appendix and shares innervation w/ kidney

Impact of Wounds or TART upon Uterus

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

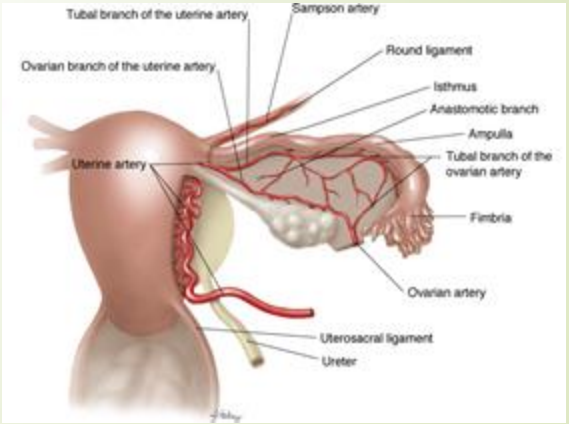
Arterial Supply: Uterine (from Internal Iliac) and Ovarian arteries

Venous Drainage: Uterine venous plexus
Connected with Superior rectal v.
Forms Portal systemic anastomosis

Lymph Drainage: 3 main routes: Aortic, Internal & External Iliac Nodes

SNS: Inferior hypogastric plexus to T11-L2 spinal levels (Inferior Mesenteric Ganglion)

PSNS: S2 – S4

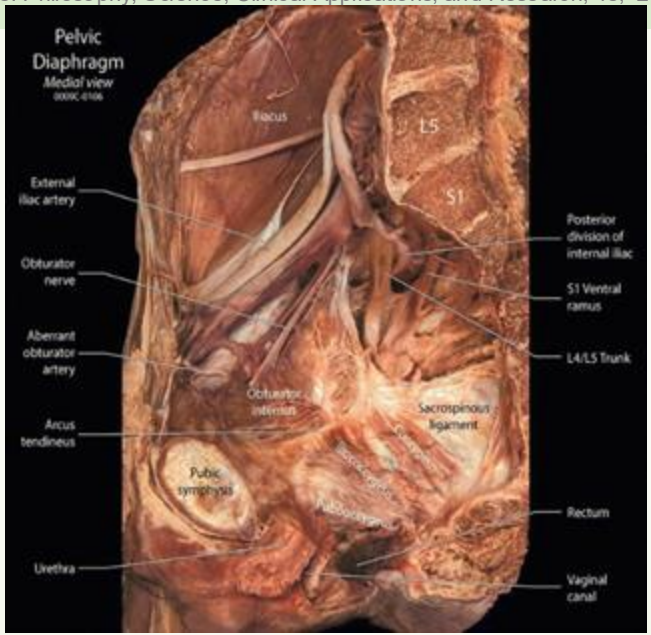


LWW Health Library

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

From: **Gynecology**

Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 4e, 2018



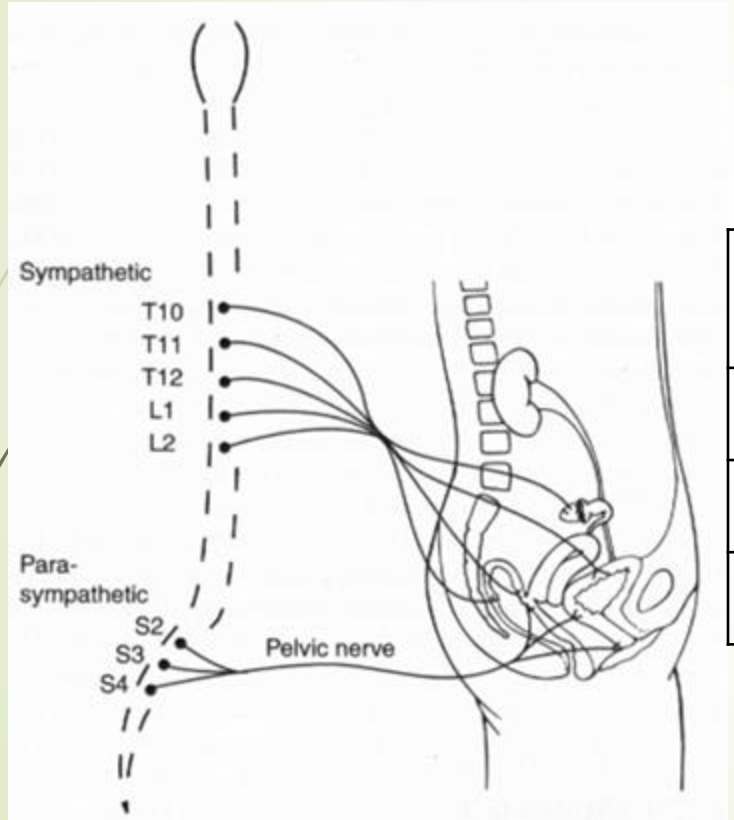
Legend:

Pelvic diaphragm, medial view. Note the levator ani muscles, the pubococcygeus and iliococcygeus. These muscles form a funnel that gives lateral and interior support to the urethra, vagina, and rectum and forms the floor of the pelvis.
(Used with permission from the Willard/Carreiro Collection.)

Date of download: 3/20/2019 Copyright © Wolters Kluwer

Innervations

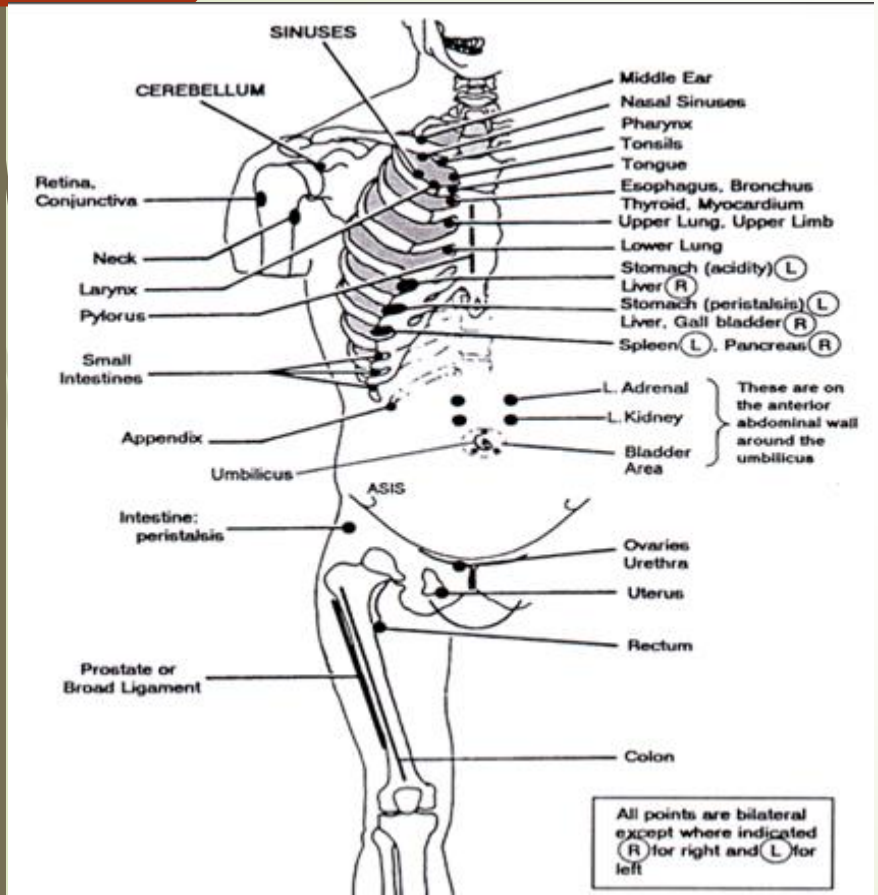
NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

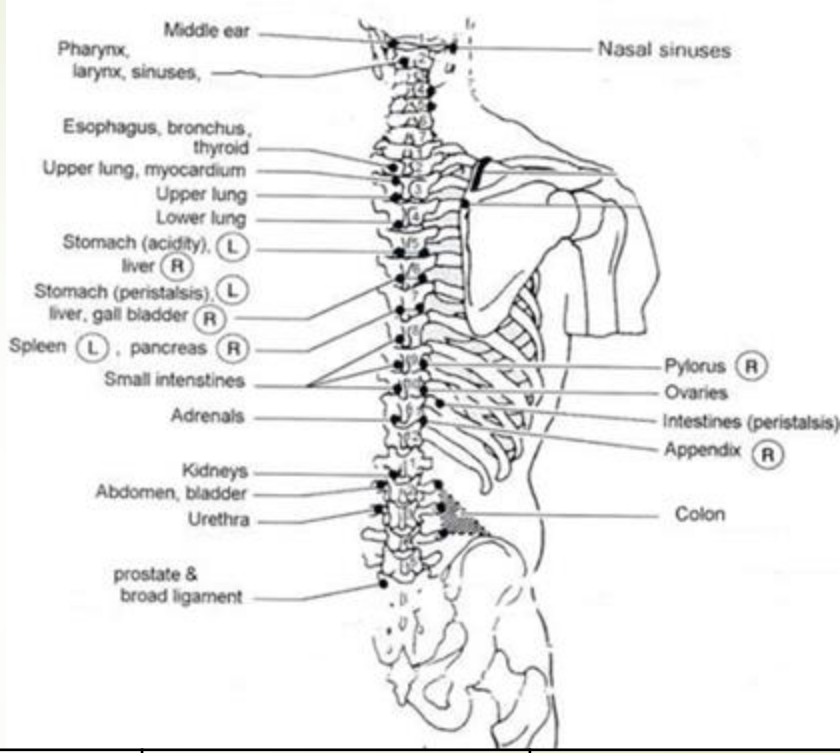


	Sympathetic (SNS)	Parasympathetic (PSNS)
Ovaries	T10-T11	Vagus (CN X)
Uterus	T11 – L2/3	S2 – S4
Cervix	T12 – L2/3	S2 – S4

Source: Kuchera

Chapman's Points*





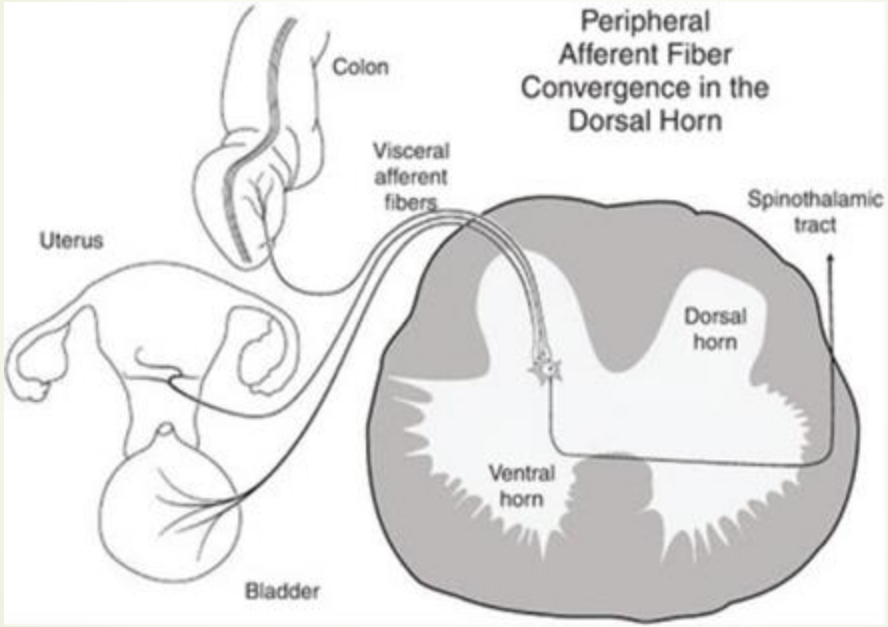
Organ	Anterior	Posterior
Ovaries	Pubic Tubercle	T10 TP
Uterus	Inferior pubic rami	Lateral sacral base
Broad Ligament	Posterior-Lateral IT band	PSIS

LWW Health Library

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

From: **Gynecology**

Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 4e, 2018



Legend:

Afferent convergence. Afferent fibers from multiple viscera are in close proximity to each other as well as in proximity to the somatic afferents. This provides an understanding for **how viscero-visceral and viscero-somatic reflexes occur**. (Used with permission from the Willard/Carreiro Collection.)

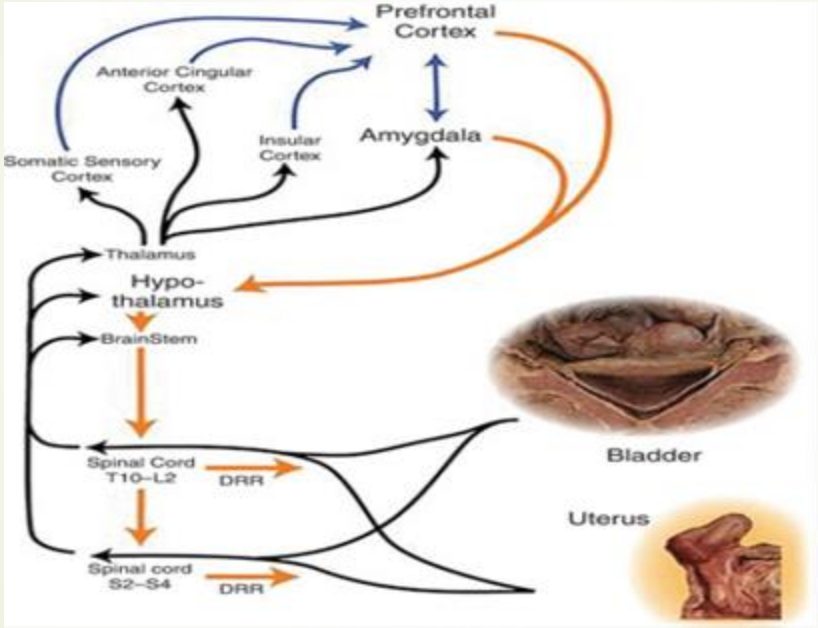
Date of download: 3/20/2019 Copyright © Wolters Kluwer

LWW Health Library

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

From: **Gynecology**

Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 4e, 2018



Legend:

Diagram showing innervation of the uterus and bladder. Note that these structures can receive dorsal root reflexes (DRRs) through the afferent fibers. This gives rise to **cross-sensitization where dysfunction of one organ can influence the surrounding organs.** This principle makes localization of visceral pain challenging.

Used with permission from the Willard/Carreiro Collection.)

Date of download: 3/20/2019 Copyright © Wolters Kluwer

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Muscles	Innervation	Function
Deep Perineal Pouch		
Striated (external) urogenital sphincter	Perineal branch of the pudendal nerve S2–S4	Voluntarily compresses the membranous urethra for continence, relaxes during micturition
Internal urethral sphincter	Sympathetic hypogastric nerves T11 to L2	Smooth muscle contraction under autonomic control
Deep transverse perineal	Perineal branch of the pudendal nerve S2–S4	Stabilizes the perineal body
Compressor urethrae	Perineal branch of the pudendal nerve S2–S4	Accessory sphincter to the urethra
Sphincter urethrovaginalis	Perineal branch of the pudendal nerve S2–S4	Accessory sphincter to the urethra, may also assist in closing the vagina
External anal sphincter	Branch of the pudendal nerve	Voluntary muscle aids in continence when constricted and relaxes for defecation
Internal anal sphincter	Autonomic nervous system	Involuntary contraction with peristalsis
Pelvic Diaphragm		Contributes to formation of the pelvic floor
Levator ani group	Direct branch of S3, S4, inferior rectal nerve, perineal nerve, S2–S4 (variations noted in cadavers)	Supports pelvic viscera, maintains the angle between the rectum and anal canal, reinforces the external anal sphincter, functions as a vaginal sphincter
Puborectalis	Direct branches of S3–S5	
Pubococcygeus	Branch of the pudendal	
Iliococcygeus	Direct Branch S3–S5	
Coccygeus	Direct branch of S3, S4	Pulls the coccyx forward after defecation
Pelvic Sidewalls		
Obturator	Obturator nerve L5–S1	Lateral rotation of extended hip, abduction of flexed hip
Piriformis	Branches L5, S1, S2	Lateral rotation of extended hip, abduction of flexed hip

Foundations of Ost Med Ed4 pg1370 Table 47B.3

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Common Gynecologic Complaints

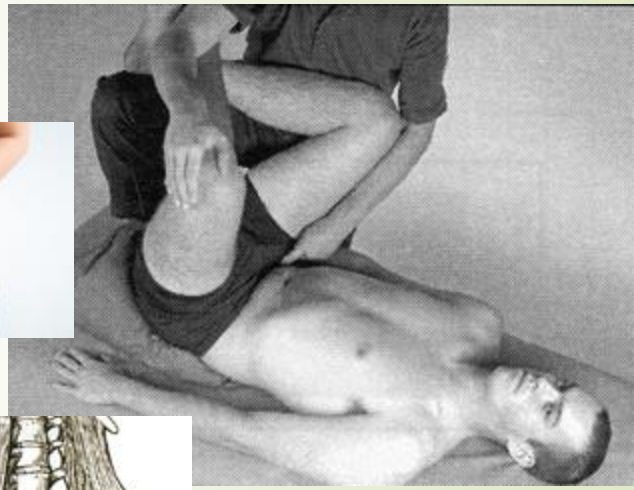
Dysmenorrhea:
Iliacus Tenderpoint Counterstrain

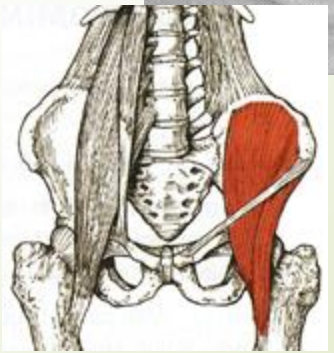
NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Treatment:

1. Standing on **same side** as tender point
2. **Marked flexion** of the hips with legs supported on doctor's thigh
3. **Marked external rotation of the legs bilaterally**
4. Think "Frog Leg"









Also known as complete rest position

Helps Decongest the pelvis

Breast: Lymphatic Drainage

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Lymphatic movement

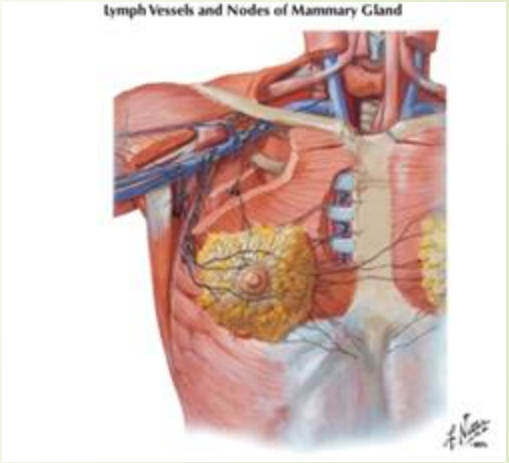
- Diaphragms
- Abdominal lifts

Scapular release

- Help swollen breasts and tenderness
- Encourage milk let down

Mastodynia

- *T1-8 (especially T4-6) Somatic dysfunction*
- *Anterior/posterior ribs*
- *Thoracic Outlet*
- *Zink treatment of breast*
- *Upper Extremity distraction*



Right Side Drainage*

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Conveys Lymph From:

1. Bronchomediastinal Trunk:

R thoracic walls

R cupola of the diaphragm and subjacent liver

R lung, bronchi and trachea

Myocardium (Pericardium drains via thoracic duct)

2. Subclavian Trunk:

R upper limb

R thoraco-abdominal wall to umbilicus anteriorly and iliac crest posteriorly

Much of R breast

3. Jugular Trunk:

R head and neck

REFERENCE text box
Grays Anatomy

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

"Your duty as a master mechanic is to know the engine is kept in perfect condition, so that there will be no functional disturbance to any nerve, or vein, or artery that supplies ad governs the skin, the fascia, the muscle the blood or any fluid that should be in free circulation to sustain life and renovate the system from deposits that would cause what we call disease."

- AT Still - Philosophy and Principles of Osteopathy pg 70





NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Osteopathic Treatment: What makes You special

Your ability to RECOGNIZE, DIAGNOSE & TREAT “Somatic dysfunctions” and to INTEGRATE this ability into the care of your patients to help optimize their health.

Prevention

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Nutrition
- Hydration
- Small Frequent Balanced meals
- Supplements
- Maintain ROM spine
- Stretches
- Exercise
- Sleep
- Stress relief
- Community building
- Self care
- Mental health
- Meditation
- Mind Body Spirit
- Work-Life-School Balance



OMT
Treat what you find.
Be sure to check:

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Transition zones/ junctions
O-A, C-T, T10-L2, L-S, SI's

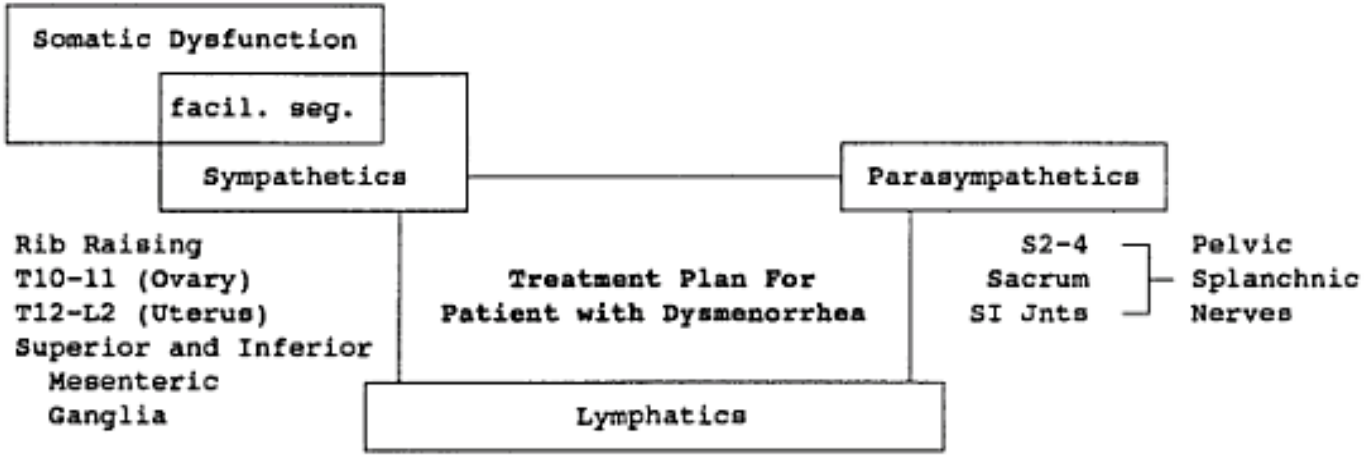
Sacral mobilization/ rocking

Lymphatic drainage
Axillary folds, diaphragms,
Thoracic Inlet and cage motion, mesenteric lifts…

Cranial
SBS, C2, OM Suture …

Gynecological Considerations (Kuchera)

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine



Question

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

A 22-year-old female presents to the office complaining of pain and cramping with her menses. She states the pelvic and back pain typically starts 1-2 days before the onset of her menses. + nausea and fatigue. On physical examination, you note a well-appearing female with a normal abdominal and pelvic examination. Structural examination will most likely reveal a sympathetic viscero-somatic reflex at:

A – OA, C2
B – T1-4
C – T11-L2
D – L3- L5
E – S2-4

Question

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

A 17-year-old overweight female presents to your office complaining of irregular and painful menses. You note on physical exam, increased acne and facial hair. Pelvic examination reveals an increased density/ areas of swelling around the ovaries. A) What do you suspect and B) Where would you most likely find a Chapman's point related to this patient's presentation?

A- Right T12 Transverse Process

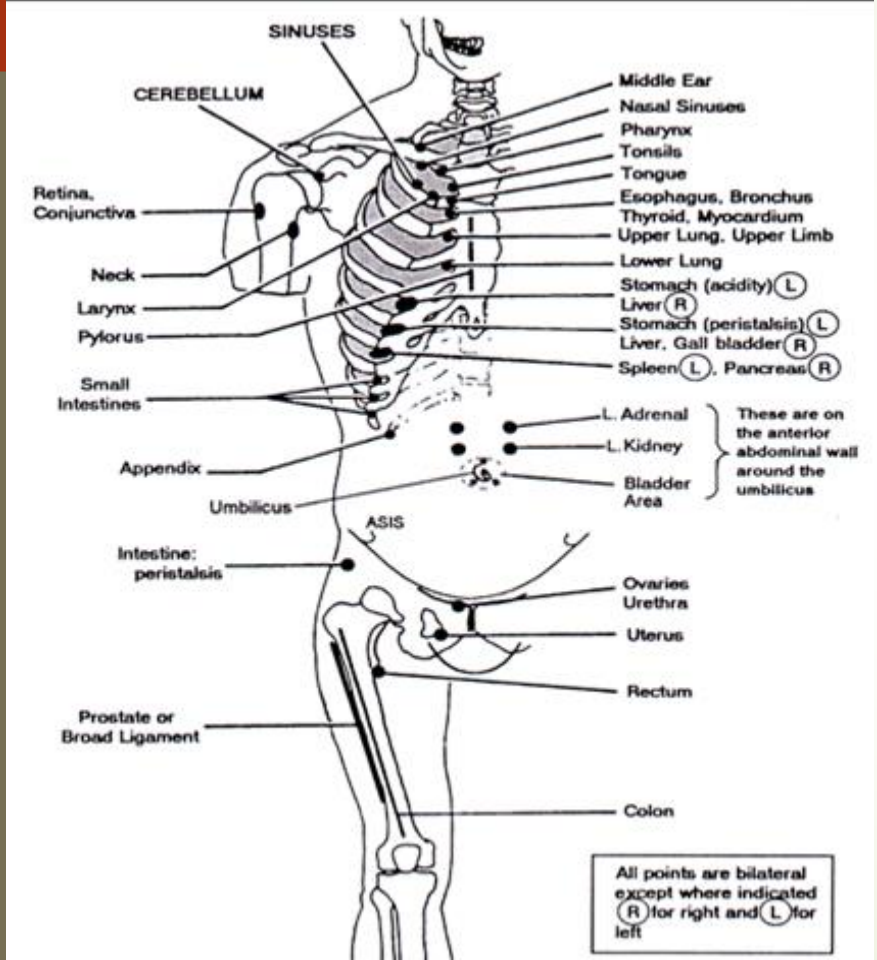
B –Lateral to the pubic symphysis on the Pubic tubercle

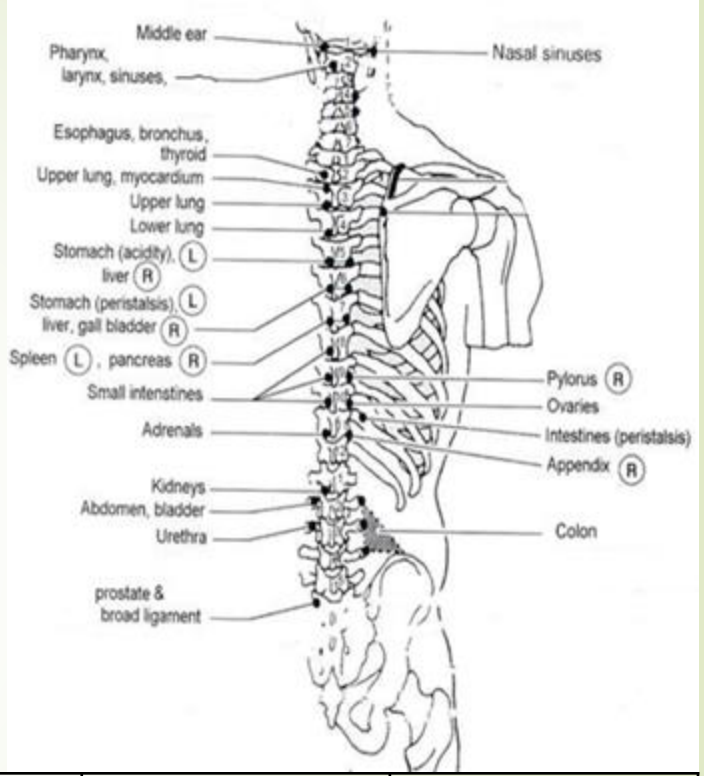
C – Upper Medial Thigh

D – Inferior pubic ramus

E – Posterior L3 Transverse Process

Chapman's Points





Organ	Anterior	Posterior
Ovaries	Pubic Tubercle	T10 TP
Uterus	Inferior pubic rami	Lateral sacral base
Broad Ligament	Posterior-Lateral IT band	PSIS

Summary Slide

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

"It matters not whether the cause is far remote or in close proximity to the uterus; we must find it, or we will be found in the antediluvian tribe of speculum cranks of all the blind female doctors' ages."

-Andrew Taylor Still
Discoverer of the Science of Osteopathy

Pg 207 on the Pelvis P&MPO



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Thank You And Good Luck!

rabusbai@nyit.edu

References:

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

REQUIRED READING:

1) The Philosophy and Mechanical Principles of Osteopathy : - Andrew Taylor Still - chp on gynecology

(Click link or cut and paste: Free Access to Text:

https://books.google.com/books?id=rBJtAAAAMAAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)

2) An Osteopathic Approach to Diagnosis and Treatment: Digiovanna et al Chp 116 3rd and/ or 4th edition :

- DSD 3rd Ed Chapter 116 Gynecologic (and OB) considerations &
- DAB 4th Ed Osteopathic Approach to Dx and Treatment Chp 116 OB & Gyn Considerations

REFERENCES:

Osteopathic Considerations in Systemic Dysfunction by Kuchera (in Genitourinary Tract Disorders p 123-44)

I invite you to fill out our Lecture Feedback Form:

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R